



Factoring with Complex Numbers

By understanding the sum of squares pattern, you can factor certain polynomials and write equations given imaginary and complex roots.

Sum of Squares

You can use complex numbers to factor a binomial that is the sum of squares.

Sum of Squares

The sum of squares pattern is

$$a^2 + b^2 = (a + bi)(a - bi).$$

Use FOIL to prove that this pattern works.

$$\begin{aligned}(a + bi)(a - bi) &= a^2 - a(bi) + (bi)a - b^2i^2 \\ &= a^2 - b^2(-1) \\ &= a^2 + b^2\end{aligned}$$

Recall that **complex conjugates** are two complex numbers of the form $a + bi$ and $a - bi$. Together, the expressions $a + bi$ and $a - bi$ form a **conjugate pair**. In the sum of squares pattern, the factors are a conjugate pair.

EXAMPLE 1

Factor $x^2 + 121$ over the complex numbers.

SOLUTION

Use the sum of squares pattern. Notice that $121 = 11^2$, so

$$x^2 + 121 = (x + 11i)(x - 11i). \blacksquare$$

EXAMPLE 2

Factor $4m^2 + 10$ over the complex numbers.

SOLUTION

Use the sum of squares pattern by rewriting $4m^2$ as $(2m)^2$ and $10 = (\sqrt{10})^2$.

$$4m^2 + 10 = (2m + i\sqrt{10})(2m - i\sqrt{10}) \quad \blacksquare$$

Creating Equations from Imaginary Roots

When the roots of a quadratic equation are opposite imaginary numbers, you can use the sum of squares pattern to write the equation that has those roots.

► **Think About It** Roots of the quadratic equation $P(x) = 0$ are solutions of the equation, which are roots of the quadratic polynomial $P(x)$. Saying that these are roots of the equation is a common shorthand.

EXAMPLE 3

The roots of a quadratic equation are $i\sqrt{8}$ and $-i\sqrt{8}$. What is the equation?

SOLUTION

If the roots of an equation are $i\sqrt{8}$ and $-i\sqrt{8}$, then the factors of the equation are $(x + i\sqrt{8})$ and $(x - i\sqrt{8})$.

By the difference of squares pattern, these factors multiply to $x^2 + (\sqrt{8})^2$ or $x^2 + 8$.

The desired equation is $x^2 + 8 = 0$, which could be written as $x^2 = -8$. $\blacksquare$

Creating Equations from Complex Conjugate Roots

When the roots of a quadratic equation are complex conjugates, you can use the sum of squares pattern to write the equation that has those roots.

How to Create a Quadratic Equation from Complex Conjugate Roots

Use these steps to create a quadratic equation from complex conjugate roots $a + bi$ and $a - bi$.

Step 1 Let the variable be x . Write the factors as $(x - (a + bi))$ and $(x - (a - bi))$ and set their product to zero.

Step 2 Rewrite the factors as $((x - a) - bi)$ and $((x - a) + bi)$.

Step 3 Multiply using the difference of squares pattern.

EXAMPLE 4

Write a quadratic equation that has roots $7 + 3i$ and $7 - 3i$.

SOLUTION

Step 1 Let the variable be x . Write the equation.

$$(x - (7 + 3i))(x - (7 - 3i)) = 0$$

Step 2 Use the associative property to rewrite the factors.

$$((x - 7) - 3i)((x - 7) + 3i) = 0$$

Step 3 Multiply the factors.

$$[(x - 7) - 3i][(x - 7) + 3i] = 0$$

$$[(x - 7)^2 - 9i^2] = 0$$

$$[(x^2 - 14x + 49) + 9] = 0$$

$$x^2 - 14x + 58 = 0$$

► **Remember** The associative property of addition states that

$$(a + b) + c = a + (b + c).$$

CHECK

Use the quadratic formula to solve the equation.

$$\begin{aligned}x &= \frac{14 \pm \sqrt{(14)^2 - 4 \cdot 1 \cdot 58}}{2 \cdot 1} \\&= \frac{14 \pm \sqrt{196 - 232}}{2} \\&= \frac{14 \pm \sqrt{-36}}{2} \\&= \frac{14 \pm 6i}{2} = 7 \pm 3i \checkmark\end{aligned}$$

The roots found using the quadratic formula are the given roots. ■

► **Think About It** When you find complex (but not real) roots using the quadratic formula, they always are a conjugate pair.